TWO BOUNDED FINITELY PROJECTIVE REPRESENTATIVES OF ONE PERFECT OBJECT AGREE AFTER LOCALIZATION UP TO TRIVIAL SUMMANDS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span includes the author's exclamation mark in 'there is an isomorphism (!) of complexes' and the phrase 'the form the form', both delivered verbatim and not corrected. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- A localization is the filtered colimit of the localizations at the powers of the elements outside the prime. invoked at proof lines 19645-19647. Descends the isomorphism over R_p to an isomorphism over R_f for some f.
- Uniqueness of a minimal complex representing a perfect object, and the lifting argument from the residue field. invoked at proof lines 19649-19651. Supplies the local case of the comparison.
- Every bounded complex of finite projective modules is a direct sum of a trivial complex and a minimal one. invoked at proof lines 19652-19653. Produces the trivial summands P^bullet and Q^bullet appearing in the conclusion.

Source reference. Original lemma 00MT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 835–835. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00CR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 471–471. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 19548: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19548–19548. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let $\mathfrak{p} \subset R$ be a prime. Let $M^\bullet$ and $N^\bullet$ be bounded complexes of finite projective R-modules representing the same object of $D(R)$. Then there exists an $f \in R$, $f \notin \mathfrak{p}$ such that there is an isomorphism (!) of complexes*

$$M_f^\bullet \oplus P^\bullet \cong N_f^\bullet \oplus Q^\bullet$$

where $P^\bullet$ and $Q^\bullet$ are finite direct sums of trivial complexes, i.e., complexes of the form the form $\ldots \to 0 \to R_f \xrightarrow{1} R_f \to 0 \to \ldots$ (placed in arbitrary degrees).

Proof. If we have an isomorphism of the type described over the localization $R_\mathfrak{p}$, then using that $R_\mathfrak{p} = \operatorname{colim} R_f$ (Algebra, Lemma 00CR) we can descend the isomorphism to an isomorphism over R_f for some f. Thus we may assume R is local and $\mathfrak{p}$ is the maximal ideal. In this case the result follows from the uniqueness of a "minimal" complex representing a perfect object, see Lemma 19548, and the fact that any complex is a direct sum of a trivial complex and a minimal one (Algebra, Lemma 00MT). □